

Summary of Developed Files

1-Android

ChatBot	
chat.py	This file handles the natural language understanding of the chatbot
Integrating_Chatbot.pdf	Documentation for the Chatbot integration to android image
main.py	A Kivy-based GUI app for the Chatbot
final_approach.png	Picture for final apk approach
Old_approach.png	Picture for old apk approach

OBD	
main.py	A Kivy-based GUI app for the Obd
OBD_App_Implementation.pdf	Documentation for Obd app integration to android image
buildozer.spec	Spec file for buildozer tool
obd-1-arm64-v8a-debug.apk	Final OBD apk that is integrated to the android image

2- Cloud

AWS_IOT CORE.pdf	Implementation of AWS IoT Core for publishing and receiving messages
Cloud Codes	
BootLoader	
AmazonRootCA3.pem	Certificate for BootLoader Thing
AmazonRootCA1.pem	Certificate for BootLoader Thing
BL_Publish	Binary outputted from compiling the BootLoader.cpp file
BootLoader.cpp	C++ code that establishes an MQTT connection with AWS IoT Core, allowing the device to publish and subscribe to messages securely.
BootLoader.json	Json file contains data for Bootloader thing
certificate.pem.crt	Certificate for BootLoader Thing
private.pem.key	Certificate for BootLoader Thing
public.pem.key	Certificate for BootLoader Thing
LSM	
AmazonRootCA3.pem	Certificate for LSM Thing
AmazonRootCA1.pem	Certificate for LSM Thing
LSM_Publish	Binary outputted from compiling the LSM.cpp file
LSM.cpp	C++ code that establishes an MQTT connection with AWS IoT Core, allowing the device to publish and subscribe to messages securely.
LSM.json	Json file contains data for LSM thing
certificate.pem.crt	Certificate for LSM Thing
private.pem.key	Certificate for LSM Thing
public.pem.key	Certificate for LSM Thing
Network	

AmazonRootCA3.pem	Certificate for Network Thing
AmazonRootCA1.pem	Certificate for Network Thing
Network_publish	Binary outputted from compiling the Network.cpp file
Network.cpp	C++ code that establishes an MQTT connection with AWS IoT Core, allowing the device to publish and subscribe to messages securely.
Network.json	Json file contains data for Network thing
certificate.pem.crt	Certificate for Network Thing
private.pem.key	Certificate for Network Thing
public.pem.key	Certificate for Network Thing
OBD	
AmazonRootCA3.pem	Certificate for OBD Thing
AmazonRootCA1.pem	Certificate for OBD Thing
OBD_publish	Binary outputted from compiling the OBD.cpp file
OBD.cpp	C++ code that establishes an MQTT connection with AWS IoT Core, allowing the device to publish and subscribe to messages securely.
OBD.json	Json file contains data for OBD thing
certificate.pem.crt	Certificate for OBD Thing
private.pem.key	Certificate for OBD Thing
public.pem.key	Certificate for OBD Thing

SomeIP_1	
AmazonRootCA3.pem	Certificate for SomeIP_1 Thing
AmazonRootCA1.pem	Certificate for SomeIP_1 Thing
SIPOne_publish	Binary outputted from compiling the SIPOne.cpp file
SIPOne.cpp	C++ code that establishes an MQTT connection with AWS IoT Core, allowing the device to publish and subscribe to messages securely.
SIPOne.json	Json file contains data for SomeIP_1 thing
certificate.pem.crt	Certificate for SomeIP_1 Thing
private.pem.key	Certificate for SomeIP_1 Thing

public.pem.key	Certificate for SomeIP_1 Thing
SomeIP_2	
AmazonRootCA3.pem	Certificate for SomeIP_2 Thing
AmazonRootCA1.pem	Certificate for SomeIP_2 Thing
SIPTwo_publish	Binary outputted from compiling the SIPTwo.cpp file
SIPTwo.cpp	C++ code that establishes an MQTT connection with AWS IoT Core, allowing the device to publish and subscribe to messages securely.
SIPTwo.json	Json file contains data for SomeIP_2 thing
certificate.pem.crt	Certificate for SomeIP_2 Thing
private.pem.key	Certificate for SomeIP_2 Thing
public.pem.key	Certificate for SomeIP_2 Thing

payload_list.txt	Text file contains paths of the json files for each thing
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3- WebApp

Flask_Server	
receiver.py	Flask server app
received_jsons	Directory that holds the incoming OBD.json files

Fleet_Management	
App.py	This is the main app that serves as the central point for navigation within the Fleet Management Web App.
app_pages	
bootloader_cloud.py	This file contains python code for bootloader page
fota.py	This file contains python code for fora page
introduction.py	This file contains python code for introduction page
lsm_overview.py	This file contains python code for lsm page
network_overview.py	This file contains python code for network page
obd_overview.py	This file contains python code for final obd page
obd_overview_V0.py	This file contains python code for old obd page
someip_overview.py	This file contains python code for someip page
BootLoader	
cloud.py	A python code that extracts features from firmware, predicts malware using a trained model
features.csv	A reference file listing expected feature columns used for model input.
label_encoder.pkl	A serialized encoder that maps predicted class indices to human-readable labels.
malware_detector_model.pkl	A trained machine learning model used to classify binary files as benign or malicious.

extra files

Cloud_arch.png	Picture for the Cloud architecture in the project
FOTA.sh	Bash script that fetches binary from Cloud and burns it on stm32
IOT_Core.png	Picture for the architecture of the AWS IoT Core service
Web_App_V1.pdf	Old Documentation for the Web App
Web_App_V2.pdf	Final Documentation for the Web App